What Does a COT Release Add?

Public positioning and risk in energy futures

Recommendation: develop a tightly specified empirical study of the incremental forecasting value of a newly published positioning report. Begin with WTI crude oil and Henry Hub natural gas. Evaluate risk forecasts as the primary outcome and return predictability and trading usefulness as secondary outcomes.

The research question

After accounting for prices, trend, volatility, the futures curve, previously released positioning and inventories, does the latest public COT report improve next-week risk forecasts? Does that improvement vary with physical tightness?

Why this could become a paper

Existing research already covers COT forecasting, machine learning, publication delay, crowding, inferred speculative positions, and hedging pressure interacting with inventories. The candidate contribution is a disciplined comparison of information already inferable before publication with information added by the new disclosure, evaluated at realistic decision times. Novelty at this exact intersection remains to be established; this review does not prove priority.

Proposed abstract - research plan, not findings

This study will examine whether public positioning disclosures add useful information to energy-futures risk forecasts beyond observable market and physical conditions. We will construct an availability-aware panel of disaggregated CFTC reports, EIA inventory releases and contract-level futures prices. At each publication, we will compare forecasts using information available immediately before release with otherwise identical forecasts that include the newly disclosed report. A rolling position-inference model will distinguish predictable positioning from model-unanticipated disclosure content. The primary evaluation will use next-week lower- and upper-quantile forecast losses, with physical tightness as a prespecified source of heterogeneity. Return prediction and net economic usefulness will be secondary analyses. Publication delays, data revisions, overlapping outcomes and repeated specification searches will be handled explicitly. The study aims to identify the conditions under which public positioning improves forecasts, or to place meaningful bounds on its incremental value.

Scope and current status

This AI-assisted planning brief builds on the audit of sinhaarya04/cot-positioning-intelligence, commit 61a8012. The repository is a useful prototype; its current reported performance is not validated evidence. No research experiments or submissions were performed for this brief. Planning assumes public data first; access to licensed contract histories remains an open feasibility question.

Closest literature: where the easy claims fail

The search covered SSRN, arXiv, publisher records, author manuscripts, university repositories and official data documentation. The studies below are the most consequential comparators, not an exhaustive systematic review.

Study	What is already established	Implication for this project
Gorton, Hayashi & Rouwenhorst (2013) [1]	Inventories, basis and momentum connect physical scarcity to commodity risk premiums.	COT must add information beyond strong physical and price baselines.
Sanders, Irwin & Merrin (2009) [2]	Little forecasting evidence from positions in ten agricultural markets; substantial evidence of trend following.	Do not assume that following large traders predicts prices.
Basu & Miffre (2013) [3]	Hedging-pressure portfolios and their relation to momentum and term structure.	New z-scores or decile portfolios are insufficient novelty.
Kang, Rouwenhorst & Tang (2020; 2021) [4,5]	Slow hedging and fast liquidity demands have different return implications; excess speculative pressure relates to factor returns.	Benchmark slow/fast pressure and crowding; do not claim either invention.
Proskurin (2021) [6]	Nonlinear COT forecasting in agriculture explicitly compares publication timing. Publisher summary reports weak Friday information and hypothetical Tuesday predictability.	Timing correction plus machine learning has direct prior art.
Boos & Grob (2023) [7]	Generalized ridge uses lagged returns to explain speculative position changes; average explanatory share exceeds 40% across 23 commodities.	Position inference is an essential benchmark, not the new contribution.
Nikitopoulos, Thomas & Wang (2024) [8]	Oil volatility, slow/fast hedging pressure, inventories and market states are studied together.	A generic COT-plus-inventory risk model is too close to existing work.
Uhl (2025); Lin, Wu & Zheng (2026) [9,10]	Speculator alignment with momentum and COT-linked extreme connectedness/bubbles are already studied.	Neither generic crowding nor generic tail risk is a safe novelty claim.

Evidence limit: some studies were accessible through abstracts, institutional records or publisher summaries rather than complete articles. In particular, the Proskurin timing conclusion is from the publisher's issue introduction. Obtain the full closest papers before finalizing the manuscript's novelty statement.

A focused contribution, with alternatives ranked

1. Recommended: incremental disclosure information

Ask whether the newly released report improves an otherwise identical forecast using the prespecified public-information baseline. Then examine a small, prespecified inventory-tightness interaction. This tests whether COT adds to the chosen observable controls; it cannot establish value beyond every public signal. Positive results, or precise bounds on negligible value, could both be economically informative.

2. More ambitious: positioning before inventory news

If survey expectations and intraday prices are available, study whether previously public crowding changes the response to EIA inventory surprises. For example, test asymmetry between crowded long positions facing bearish news and crowded shorts facing bullish news. This offers a sharper event mechanism but needs credible expectation data, precise event windows and a stronger identification argument. It is a separate next-stage paper, not a small addition to the main study.

3. Supporting study: the useful life of disclosures

Compare fresh, stale and inferred positioning as reports age. Treat unusual delays as stress tests. Routine report age has little variation, and shutdowns or operational failures coincide with other shocks; they do not automatically identify the causal effect of transparency. A release-date benchmark is useful infrastructure, but release-aware datasets already exist [25].

What arXiv contributes to the methods

Research	Appropriate use here
Xu & Xie, SPCI (2022/2023) [11]	Optional sequential uncertainty calibration after the core forecast experiment. Check assumptions and performance under dependent observations.
Gibbs & Candes, adaptive conformal inference (2021) [12]	A robustness approach to changing distributions. Long-run coverage is not conditional coverage in every inventory state.
Harvey & Liu, false and missed discoveries (2020) [13]	Predeclare comparisons, record all trials, quantify power and account for multiple testing.
Glasserman & Lin, GPT look-ahead bias (2023) [14]	A warning for historical LLM backtests when model training can include later information. Keep the LLM outside the primary forecasting test.

Scientific core: a small interpretable model and a fair information comparison. A larger model is justified only by demonstrable out-of-sample improvement over these baselines, not by its architectural novelty.

The experiment: define exactly what is new

Unit: one publication event per commodity. Begin with futures-only disaggregated COT for WTI and Henry Hub. Exclude backfilled pre-publication years from the tradable evaluation. Use at least five years of available history for the main warmup; the eventual evaluation start depends on verified coverage. Three-year warmup is a sensitivity analysis, not a way to manufacture a larger test set.

Model	Information allowed at the forecast cutoff
M0: public baseline	Trailing dollar-price changes/trend, realized dollar volatility, curve slope, seasonality, released inventories, previously released COT and its age. Volume/OI only if their publication times are known.
M1: new disclosure	M0 plus current net managed-money positioning divided by report open interest. This matched-model comparison is the primary test of incremental COT information.
M2: physical-state interaction	M1 plus a single positioning-by-inventory-tightness interaction. Primary proxies: released Cushing crude stocks and Lower-48 working gas; negative inventory anomalies relative to the prior five years' same calendar week +/- two weeks, standardized using that past sample.
Interpretation exercise	Infer current reported positioning using pre-release public data; compare model-predicted positioning with the disclosed value. Use past-only cross-fitted predictions when constructing historical residual features.

Notation: q = (managed-money long - managed-money short) / report open interest. Let qhat be a prediction of q using information available immediately before release; u = q - qhat is model-unanticipated positioning. Its normalization and model choices must be fitted on past data.

Critical distinction: qhat + u equals q. Residualization does not create information, and separate coefficients can increase capacity. Use it for interpretation and matched-complexity robustness; base the headline on M1 versus M0. The residual is neither a consensus surprise nor an exogenous shock. Inventory anomalies proxy tightness; low stocks and full-storage constraints can create different risks. Estimate market effects separately and examine upper/lower inventory states without assuming one sign.

Outcome, loss and economic interpretation

Predict the next five trading sessions' contract-dollar P&L, scaled by trailing dollar-change volatility fixed before release. This avoids undefined or misleading log/percentage returns around negative WTI prices. The primary score averages 10th- and 90th-percentile pinball losses with equal weights for the two markets. Compare each model on identical events, outcomes and scale.

Report lower and upper quantiles separately as secondary outcomes: an average gain alone does not prove asymmetry. Mean-return forecasts, four-week horizons, gross long/short decompositions, 5% tails and a fixed risk-budget application are secondary. Do not make 1% tail claims from a small weekly sample.

Data credibility is a prerequisite

Required clock or field	Reason and implementation
Observation / publication / knowledge time	Store the position date separately from actual release time and the first time each value was available. Record source, retrieval timestamp, content hash, revision status and confidence in the reconstructed release time.
COT start and exceptional releases	DCOT began on 4 September 2009; history back to June 2006 was published later [17]. Do not treat that backfill as tradable before release. Reconstruct holidays and unusual delays using official announcements [18].
EIA inventory vintages	Use released values and release dates from archived reports where possible. Current history can include revisions or back-estimates. Distinguish a true first-vintage test from a latest-vintage, timestamp-correct approximation [19,20].
Contract identity and maturity	Select the nearest contract with at least ten exchange trading days to expiry at entry, using the known expiry calendar; hold that contract for the five-session target. Use the next eligible maturity for curve slope. Economic portfolios must account for closing, reopening and roll costs. COT aggregates maturities.
Execution benchmark	Use a price strictly after publication. Normal Friday COT publication is after the usual CL/NG settlement window, so the first later settlement often is Monday [21]. Settlement is a benchmark, not a guaranteed fill; report costs and execution sensitivity.
Availability uncertainty	Main sample should favor verified releases and recoverable vintages. Report counts and exclusions. Add conservative-delay and latest-vintage sensitivity analyses instead of silently imputing certainty.

Information timing example

Tuesday positions observed -> Friday report published -> next admissible pricing benchmark -> five-session outcome. All M0 controls are frozen immediately before the Friday disclosure; M1 receives only the additional report. Both models use the same delayed price benchmark. This estimates predictive usefulness with that delay, not the immediate price impact of publication.

Feasibility gate

CFTC and EIA provide public inputs. A clean, sufficiently long contract-level price and curve history is the main unresolved data dependency. CME historical products have licensing conditions [22]. Yahoo continuous bars can support a prototype, but do not by themselves establish contract rolls, executable fills or a redistributable term-structure panel. Confirm available university/vendor access before promising a full economic backtest.

Validation that can support a research claim

Chronological evaluation

Fit expanding or fixed rolling models on completed historical outcomes only, with annual calendar blocks for outer evaluation and past-only inner tuning. Keep both markets in the same temporal fold. Fit scaling, seasonal inventory references, winsorization, quantile thresholds and position-inference models inside each training window. Purge labels that extend past the next forecast cutoff.

The historical data have already been explored by this project, so describe this as pseudo-out-of-sample evidence. Freeze the protocol and archive future forecasts before outcomes occur to build a genuinely prospective record. A few months of prospective results are an operational check, not a well-powered proof of forecast skill.

Baselines, inference and stopping rules

Use historical quantiles, volatility-scaled historical quantiles and a heavy-tailed GARCH/EGARCH benchmark, followed by parsimonious regularized quantile regressions on M0/M1/M2. One small boosted-tree model may be a robustness exercise. Match tuning opportunities between information sets.

Estimate loss differences with common-date block resampling that preserves serial dependence, cross-market shocks and any overlapping holding periods. Two markets do not justify commodity-cluster asymptotics. Conditional forecast evaluation has an established econometric foundation [15]. Correct the small prespecified family of secondary tests; report every attempted specification in a trial ledger [13].

Before evaluation, set an economically meaningful minimum improvement using training-period simulations and the intended risk-budget use. A non-significant result is not proof of zero value. Report confidence intervals and whether they exclude the minimum useful effect. If intervals remain wide, the honest conclusion is limited power.

Predeclared comparison	Interpretation
M1 versus M0: primary	Does the latest report improve the chosen risk forecast beyond already public inputs?
M2 versus M1: secondary	Does a prespecified physical-tightness interaction improve forecasts?
Raw versus inferred/residual positioning	How much of the apparent signal is predictable from public data? Descriptive decomposition, not causal identification.
Availability, vintage and execution ablations	How sensitive is the result to data timing, revisions and practical delays? These are diagnostics, not new trading results.

Paper outputs: coverage/availability audit; benchmark loss table with intervals; cumulative loss-difference chart; separate tail calibration by predeclared state; economic sensitivity table; full experiment ledger. No performance values should enter the manuscript until reproduced by the final pipeline.

Translate the audit into a research codebase

The local audit covered all 11 tracked files, including nine Python files. It reproduced defects rather than validating historical returns. The changes below are prerequisites for evidence, not claims of academic contribution. The audited source remains unchanged.

Current file or component	Required improvement and verification
src/ingest_cftc.py	Correct field mappings; reject missing required columns instead of replacing them with zeros. Validate category sums, unique report keys and known official rows. Record report type and market code.
src/ingest_prices.py	Remove the exposed key from code and rotate it through the provider account. Store daily contract histories and EIA release/vintage metadata. Validate pagination and full date coverage.
src/signals.py and root signals.py	Consolidate duplicate implementations. Replace nearest-week joins with backward availability joins. Construct seasonal inventory tightness using training history only.
src/backtest.py	Replace ambiguous weekly Yahoo timestamps and nearby-price lookup with explicit event/entry/exit timestamps. Eliminate full-sample deciles. Separate forecast loss from signed strategy P&L, costs and portfolio metrics.
src/db.py and ingestion writes	Use transactional, idempotent updates; preserve immutable raw data and provenance. Fail clearly on missing/empty datasets. Keep revisions auditable.
src/llm_anomaly.py	Remove the LLM directional score from the scientific core. Retain optional evidence-linked summaries with schema, range and market validation; freeze model/prompt versions for any separate evaluation.
src/dashboard.py and README.md	Label observation, release and forecast dates; display sample size, uncertainty and vintage quality. Replace unsupported performance statements with regenerated results.
New research infrastructure	Dependency lockfile; data dictionary; release calendar; experiment configuration; trial ledger; meaningful leakage/roll/schema tests; one-command generation of tables and figures.

A conditional 10-12 week plan

Weeks 1-2: secure price/curve access, audit vintages, repair ingestion and obtain the closest full papers. **Weeks 3-4:** reproduce feasible published benchmarks, implement minimal models and freeze the protocol. **Weeks 5-7:** run chronological experiments and inference. **Weeks 8-9:** robustness, economic interpretation and independent review. **Weeks 10-12:** write the complete manuscript, package reproducibility materials and prepare submission. Data access or weak statistical power may require more time; prospective validation continues beyond this schedule.

Publication route: choose after seeing the evidence

Destination	Role and realistic fit
SSRN [26]	A complete eligible working paper first. Posting is not peer review. Current guidelines require rigorous methodology/original findings and a full English PDF; AI use must be disclosed in the abstract and PDF. A proposal or framework alone is not the intended submission.
arXiv q-fin.ST [27]	Optional quantitative-finance preprint route for the econometric study. Endorsement and moderation may apply. It also does not constitute journal peer review.
Applied Finance Letters [28]	Plausible focused-paper route if one clear empirical finding survives the checks. Official guidance: double-blind review, no submission/publication/access fees, and 3,500 words including abstract, body and footnotes. Target a regular issue.
Journal of Commodity Markets [29]	Best topical fit for a deeper full paper on commodity forecasting, inventories and risk. Requires an original economic result and robust evidence beyond repairing a prototype.
Journal of Futures Markets [30]	Strong fit for derivatives risk/forecasting once incremental information and economic interpretation are convincingly established.
Energy Economics [31]	Stretch target if a substantive energy-market mechanism and replicable extension emerge. A stronger model leaderboard alone is unlikely to carry the contribution.

Venue fit is a research judgment, not an acceptance forecast. Fees for the three larger journals were not verified. Recheck current author guidance, eligibility and preprint policies before uploading, and submit to only one journal at a time. SSRN eligibility can depend on content and author status; this brief does not establish eligibility.

What a submission-ready package must contain

A complete paper with actual estimates, uncertainty and limitations; a data dictionary and availability audit; the frozen protocol and experiment ledger; reproducible code and environment; legally shareable data or acquisition instructions; a blinded manuscript where required; author/funding/conflict details and an accurate AI-use disclosure. Request academic feedback on the closest-paper comparison before submitting.

Go / revise / stop

Go: independently reproduced incremental value or precise, economically meaningful bounds, with timing/vintage robustness. **Revise:** useful evidence but insufficient mechanism or generalization; aim for a focused note. **Stop expanding the model:** poor data or imprecise results. Publish an honest technical report or improve the data rather than searching markets until a positive result appears.

Recommended sequence: complete the empirical study, circulate for feedback, post an eligible preprint if appropriate, then choose AFL for a concise result or JCM for a full contribution. No paper has been submitted or published as part of this task.

References and research trail

Sources checked 15 October 2026. Each source link leads to a primary publisher, author, institution or official documentation. This is an annotated proposal bibliography, not a claim that every full text was accessible.

[1] Gorton, G. B., Hayashi, F. & Rouwenhorst, K. G. (2013). The Fundamentals of Commodity Futures Returns. Review of Finance 17(1), 35-105. Earlier NBER working paper 13249. Source.

[2] Sanders, D. R., Irwin, S. H. & Merrin, R. P. (2009). Smart Money: The Forecasting Ability of CFTC Large Traders in Agricultural Futures Markets. JARE 34(2), 276-296. Author-institution record. Source.

[3] Basu, D. & Miffre, J. (2013). Capturing the Risk Premium of Commodity Futures: The Role of Hedging Pressure. Journal of Banking & Finance 37(7), 2652-2664. Source.

[4] Kang, W., Rouwenhorst, K. G. & Tang, K. (2020). A Tale of Two Premiums: The Role of Hedgers and Speculators in Commodity Futures Markets. Journal of Finance 75(1), 377-417. Institutional manuscript. Source.

[5] Kang, W., Rouwenhorst, K. G. & Tang, K. (2021). Crowding and Factor Returns. Working paper, 13 March. University-hosted manuscript. Source.

[6] Proskurin, O. (2021). Does the CFTC Report Have Predictive Power: Machine Learning Approach. Journal of Financial Data Science 3(3), 139-151; DOI 10.3905/jfds.2021.1.065. Timing findings verified from the publisher's issue introduction; full article access restricted. Source.

[7] Boos, D. & Grob, L. (2023). Tracking Speculative Trading. Journal of Financial Markets 64, 100774. Institutional record and manuscript. Source.

[8] Nikitopoulos, C. S., Thomas, A. C. & Wang, J. (2024). Hedging Pressure and Oil Volatility: Insurance versus Liquidity Demands. Journal of Futures Markets 44(2), 252-280. Source.

[9] Uhl, B. (2025). Speculators and Time Series Momentum in Commodity Futures Markets. Review of Financial Economics 43(2), 213-230. Publisher abstract. Source.

[10] Lin, B., Wu, J. & Zheng, Q. (2026). Speculative Pressure, Extreme Connectedness, and Price Bubbles in Oil and Agricultural Commodity Markets. Journal of Commodity Markets 42, 100561. Publisher abstract. Source.

[11] Xu, C. & Xie, Y. (2022; revised 2023). Sequential Predictive Conformal Inference for Time Series. arXiv:2212.03463; ICML 2023. Source.

[12] Gibbs, I. & Candes, E. (2021). Adaptive Conformal Inference Under Distribution Shift. arXiv:2106.00170; NeurIPS 2021. Source.

Methods, data and publication sources

[13] Harvey, C. R. & Liu, Y. (2020). False (and Missed) Discoveries in Financial Economics. arXiv:2006.04269. Source.

[14] Glasserman, P. & Lin, C. (2023). Assessing Look-Ahead Bias in Stock Return Predictions Generated By GPT Sentiment Analysis. arXiv:2309.17322. Source.

[15] Giacomini, R. & White, H. (2006). Tests of Conditional Predictive Ability. Econometrica 74(6), 1545-1578. Source.

[16] Robe, M. A. & Roberts, J. S. (2023 working paper). Four Commitments of Traders Report Puzzles, Revisited: Answers from Grains and Oilseeds Futures Markets. Additional context on aggregate trader-category limitations. Source.

[17] CFTC (2009). Launch of disaggregated COT reports and subsequent historical-data release. Official historical release announcement. Source.

[18] CFTC. Historical Special Announcements, including exceptional delays and corrections. See also COT FAQ and release schedule. Source.

[19] EIA. Weekly Petroleum Status Report archive: release dates and data-ending dates. Source.

[20] EIA. Weekly Natural Gas Storage Report methodology; back-estimation and revision context. Source.

[21] CME Group. NYMEX WTI Crude Oil, Natural Gas, Heating Oil and RBOB Gasoline Futures Daily Settlement Procedure. Historical conventions must be verified for the sample. Source.

[22] CME Group. Continuous Price Series; contract identifiers, settlements and roll metadata. Access and redistribution require license review. Source.

[23] CFTC. COT reports, explanatory material and FAQ. Additional source for publication-time and classification limitations. Source.

[24] EIA. API technical documentation, including pagination limits. Additional implementation reference. Source.

[25] ZipLime. Commitments-of-traders dataset card with release_at, knowledge_date and provenance tiers. Creator claims require independent validation; cited only as existing tooling. Source.

[26] SSRN Support. Submission guidelines, updated 3 August 2026; full paper, eligibility and AI-disclosure requirements. Source.

[27] arXiv. Category taxonomy (q-fin.ST). See also help/endorsement and help/moderation for submission checks. Source.

[28] Applied Finance Letters. Author instructions: length and review requirements. No-fee policy verified on the journal's About page. Source.

[29] Elsevier. Journal of Commodity Markets: official scope. Source.

[30] Wiley. Journal of Futures Markets: official journal site and author guidance. Source.

[31] Elsevier. Energy Economics: official scope and replication emphasis. Source.